

Notes: Celentano & Rempel (RES, 2026)

Public Listing Choice with Persistent Hidden Information

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1 Overview

1.1 Main question

Why do some firms remain public while others stay private or go private—and why has the propensity to be public declined at the same time that public CEO compensation became more incentive-based?

1.2 Core mechanism

A firm’s cash flows contain a **persistent hidden component** that outside (public) investors cannot verify. To induce truthful disclosure, public investors must design incentive contracts. Because CEOs are risk-averse (and can save/borrow), these incentives are costly.

1.3 Competition between two financing technologies

- **Public finance (P)**: deep funds but cannot observe the hidden component z_t ; must use incentive pay.
- **Private finance (S)**: can monitor z_t by paying a cost ν ; but private capital is scarce/limited.

1.4 Equilibrium implication

Competition plus scarcity produces **sorting**:

- *Information-intensive* firms (high hidden volatility and/or high exposure to hidden cash flows) are disproportionately **privately financed**.
- *Transparent / tangible* firms (high τ) and/or firms whose hidden component is *predictable* (low σ_z or low persistence) are more likely **public**.
- Public CEOs face higher incentive risk precisely in the set of firms that remains public.

1.5 Empirical and structural contributions

- Reduced-form evidence using U.S. private and public firm data supports the joint sorting prediction in *tangibility* and *volatility*.
- CEO pay regressions show pay is higher and more incentive-based when intangibility is higher and cash flows are riskier.
- A structural GMM estimation identifies the persistence and volatility of the hidden cash-flow component using (i) listing choice and (ii) CEO pay-performance sensitivity.

2 Model environment (cash flows, information, preferences)

2.1 Cash-flow decomposition and tangibility

Total (per-unit) cash flow is

$$y_t \equiv \tau x_t + (1 - \tau)z_t, \quad \tau \in [0, 1]. \quad (1)$$

- x_t : “tangible” and publicly observable.
- z_t : “intangible” and privately observed by the CEO.
- τ : **tangibility share**. Higher τ means less exposure to private information.

2.2 Persistence: OU dynamics (continuous-time AR(1))

Each component follows an Ornstein–Uhlenbeck process:

$$di_t = \left(\mu^i - \frac{i_t}{\lambda^i} \right) dt + \sigma^i dW_t^i, \quad i \in \{x, z\}. \quad (2)$$

The stationary mean is $\mathbb{E}[i_t] = \lambda^i \mu^i$. A larger λ^i implies a more persistent process (mean-reversion rate $1/\lambda^i$).

2.3 Information structure

x_t is observable and contractible. z_t is private to the CEO. Thus the key friction is **hidden information about a persistent state**.

2.4 Preferences and outside option

The CEO has CARA utility $u(c) = -\exp(-\psi c)$ with risk aversion $\psi > 0$. The CEO can save/borrow at a risk-free rate r . The CEO's outside option corresponds to promised utility q .

3 Two financing technologies and the information premium

3.1 Investor payoff

Financing requires one unit of funding at $t = 0$. If financed by type $f \in \{P, S\}$, the investor chooses a compensation contract $\{\omega_t^f\}_{t \geq 0}$. The investor's expected discounted return is

$$R^f(q_0, \theta) = \mathbb{E} \left[\int_0^\infty e^{-\rho t} (y_t - \omega_t^f(q_0, \theta)) dt \mid \theta \right] - \nu \mathbf{1}\{f = S\} - 1, \quad (3)$$

where ν is the lifetime monitoring cost for private financing.

3.2 Information premium and the private advantage wedge

Let $\pi(q_0, \theta)$ be the information rent (extra cost) required under public finance. Then

$$R^S(q_0, \theta) - R^P(q_0, \theta) = \pi(q_0, \theta) - \nu. \quad (4)$$

Interpretation: private finance is strictly more valuable when $\pi > \nu$ (monitoring dominates incentives).

4 Optimal contracting and the cost of persistent hidden information (Theorem 1)

4.1 Information premium formula

Theorem 1. The information premium is independent of q_0 and equals

$$\pi(\theta) = \frac{\psi}{2} \left(\frac{(1 - \tau)\sigma^z}{\rho + (\lambda^z)^{-1}} \right)^2. \quad (5)$$

Why persistence matters. Holding ρ fixed, increasing λ^z decreases $(\lambda^z)^{-1}$, reducing the denominator and increasing π . Intuitively, when hidden information is persistent, incentive exposure must discipline a long-lived state, which is expensive under risk aversion.

4.2 Contract form (public vs private)

Theorem 1 (contract representation). Public-funding CEO compensation can be written as

$$\omega_t^P(q_0, \theta) = \omega_{\text{fix}}(q_0) + \omega_{\text{gro}}(\theta) t + \omega_{\text{per}}(\theta) W_t^z, \quad (6)$$

where

$$\omega_{\text{fix}}(q_0) = -\frac{1}{\psi} \log(-\rho q_0), \quad (7)$$

$$\omega_{\text{gro}}(\theta) = \rho^2 \pi(\theta), \quad (8)$$

$$\omega_{\text{per}}(\theta) = \rho \sqrt{\frac{2}{\psi} \pi(\theta)}. \quad (9)$$

Private-funding compensation is a constant salary:

$$\omega_t^S(q_0, \theta) = \omega_{\text{fix}}(q_0). \quad (10)$$

Interpretation. Public investors cannot observe z_t , so they tie compensation to the innovation W_t^z (a reduced-form representation of exposure to hidden shocks). The growth term compensates the CEO over time for bearing incentive risk when the CEO can privately save/borrow. Private monitoring replaces incentive exposure, so private CEOs are paid more “fixed”.

5 Financing equilibrium and firm sorting (Theorem 2 and Corollary 1)

5.1 Return decomposition and individual rationality

Investor returns can be decomposed as

$$R^f(q_0, \theta) = \Upsilon(\mu) - \Lambda^f(\pi) - X(q_0), \quad (11)$$

where

$$\Upsilon(\mu) \equiv \frac{\mu - \omega^S(q)}{\rho} - 1, \quad (12)$$

$$\Lambda^S(\pi) = \nu, \quad \Lambda^P(\pi) = \pi, \quad (13)$$

$$X(q_0) \equiv \frac{\omega^S(q_0) - \omega^S(q)}{\rho}. \quad (14)$$

Financing requires individual rationality at the outside option q :

$$\Upsilon(\mu) - \Lambda^f(\pi) \geq 0. \quad (15)$$

5.2 Sorting with scarce private funds

Private capital has limited budget B . Equilibrium is characterized by two cutoffs: a profitability cutoff $\underline{\mu}$ and an information-premium cutoff $\underline{\pi}$.

- Firms with very low μ are unfinanced.
- Among financeable firms, private finance targets those with sufficiently large π (high monitoring value) subject to budget.
- Remaining firms are publicly financed.

This sorting is depicted in Figure 1(a).

5.3 Corollary 1: Sorting in (τ, σ^z)

Conditional on $\mu \geq \underline{\mu}$ and holding persistence fixed, public listing is equivalent to a volatility cutoff:

$$\sigma^z(\tau) = \frac{1}{1-\tau} \sqrt{\frac{2\pi}{\psi}} (\rho + (\lambda^z)^{-1}). \quad (16)$$

Interpretation: more intangible firms (τ low) need very low hidden volatility σ^z to remain public.

6 Endogenous private equity supply and matching (Theorem 3)

Households with heterogeneous monitoring costs $\nu \sim G_\nu$ can become private financiers. The equilibrium has:

- a cutoff $\bar{\nu}$ such that households with $\nu \leq \bar{\nu}$ enter,
- aggregate private funds $B = M G_\nu(\bar{\nu})$,
- negative assortative matching: firms with higher monitoring value match with lower- ν households.

This makes the “PE premium” endogenous.

7 Comparative statics and secular trends (Table 1)

Table 1 reports sign predictions (SR vs LR) for four trends. Below is the intuition for each row.

7.1 Trend 1: rising intangibility

From (5), lowering τ raises π .

- **SR:** more firms become information intensive $\Rightarrow$ fewer public listings; public CEO pay rises and becomes more incentive-based.
- **LR:** higher π raises returns to monitoring $\Rightarrow$ more private-capital entry ($B \uparrow$), partially offsetting the SR effect.

7.2 Trend 2: PE deregulation / cheaper monitoring

A shift in G_ν toward lower ν raises private-capital supply. This expands private financing and reduces the public set.

7.3 Trend 3: lack of ideas

A fall in profitability μ shrinks the set of financeable firms and depresses output. Listing effects depend on which firms exit (selection matters).

7.4 Trend 4: higher unproductive disclosure costs

Higher public-market compliance/disclosure costs shift firms away from public markets without reducing π . The model predicts fewer listings and lower output.

TABLE 1
Comparative statics

	Listing Propensity (L)		PE Funds (B)		Public CEO pay (Ω)		PE Premium (Π)		Aggregate Output (O)	
	SR	LR	SR	LR	SR	LR	SR	LR	SR	LR
(i) Intangibility ($\uparrow G_x$)	-	-	·	+	+	?	+	?	-	-
(ii) PE Deregulation ($\downarrow G_v$)	·	-	·	+	·	-	·	?	·	+
(iii) Lack of Ideas ($\downarrow G_\mu$)	?	?	·	-	?	?	?	?	-	-
(iv) Disclosure Costs ($\uparrow \zeta$)	-	?	·	+	?	?	+	?	-	-

Notes: This table presents the theoretical comparative static general equilibrium predictions for various economic aggregates across competing theories of secular trends introduced in Theorem 4. Specifically, we study the impact of increasing (i) intangibility, (ii) PE deregulation, (iii) lack of ideas, and (iv) disclosure costs of being public. The considered economic aggregates are the listing propensity (L), the supply of PE funds (B), the average public CEO compensation (Ω), the average PE premium (Π), and the aggregate output (O). For all the experiments, short-run (SR) and long-run (LR) consequences for each aggregate are considered. We define the SR as the case in which the monitoring cost cutoff $\bar{v}$ is fixed and so is the supply of PE funds, B , while in the LR case we allow the cutoff to adjust. The symbol + (-) denote a rise (fall) relative to the initial level, · indicates not applicable, while ? means that an effect is taken to be ambiguous. See Appendix A.3 for details.

TABLE 2
Descriptive statistics

	Public				Private			
	Mean	Median	Std	N	Mean	Median	Std	N
Firm Characteristics								
Book Assets	3,184.06	358.52	15,077.42	74,409	973.74	34.52	5,337.31	45,482
Firm Age	34.09	25.00	25.44	74,409	20.31	12.00	21.74	45,482
Tangibility	0.64	0.64	0.20	74,409	0.61	0.63	0.25	45,482
Volatility	0.03	0.02	0.04	65,523	0.06	0.03	0.08	25,689
Profitability	0.07	0.09	0.12	74,272	0.01	0.05	0.18	43,982
Sales	2,837.70	342.40	13,386.80	74,409	713.05	33.03	3,422.82	45,482
PPEGT	0.46	0.39	0.31	72,805	0.46	0.38	0.34	41,608
Capital Expenditure	0.14	0.10	0.14	72,614	0.19	0.11	0.22	40,117
R&D	0.11	0.08	0.11	28,061	0.17	0.11	0.17	12,682
SG&A	0.27	0.21	0.25	73,386	0.45	0.28	0.53	44,729
Goodwill	0.16	0.12	0.15	42,369	0.22	0.17	0.19	16,007
CEO Pay Characteristics								
CEO Pay	4.18	1.66	9.52	45,329	1.20	0.37	4.07	19,594
Fixed Share	0.52	0.44	0.34	45,329	0.79	0.96	0.29	19,594
Incentive Share	0.61	0.67	0.26	32,816	0.45	0.43	0.28	6,754
Equity Share	0.51	0.53	0.24	30,768	0.41	0.37	0.29	5,588

Notes: This table reports descriptive statistics for firm characteristics from 1993 to 2016 as well as CEO compensation characteristics from 2001 to 2016 for companies in our sample. The table is divided between observations related to public and private firms. For each variable we report its mean, median and standard deviation. All nominal values are adjusted to 2016 U.S. dollars. We annually winsorize scaled variables without clear upper or lower bounds at the 1% and 99% level.

8 Empirics: measures and descriptive evidence (Table 2)

8.1 Key measures

- **Tangibility proxy:** the paper uses a balance-sheet based measure of tangibility, consistent with the model's τ .
- **Volatility proxy:** three-year standard deviation of EBITDA/Assets.
- **CEO pay:** total pay and its decomposition into fixed, incentive, and equity components.

8.2 Table 2 interpretation (means)

Table 2 contrasts public vs private firms. Key mean differences (public minus private):

- **Size and age:** public firms are much larger (mean book assets 3184 vs 974) and older (34 vs 20 years).
- **Tangibility and volatility:** public firms are slightly more tangible (0.64 vs 0.61) and much less volatile (0.03 vs 0.06).
- **Spending composition:** private firms spend more on CAPEX, R&D, and SG&A as shares of assets.
- **CEO pay levels:** public CEOs earn about 4.18 vs 1.20 (units as reported), consistent with stronger incentive requirements.
- **CEO pay composition:** private CEOs have a higher fixed-pay share (0.79 vs 0.52) and a lower incentive share (0.45 vs 0.61).

These patterns line up with the model: private monitoring substitutes for incentive exposure.

9 Reduced-form tests: Tables 3–5 and Figures 1–5

9.1 Listing regressions (Table 3) and industry sorting (Figure 2)

Empirical model (linear probability). The baseline specification is a linear probability model:

$$\text{Public}_{it} = \alpha + \beta_\tau \text{Tangibility}_{i,t-1} + \beta_\sigma \text{Volatility}_{i,t-1} + \gamma' X_{i,t-1} + \text{FE} + \varepsilon_{it}, \quad (17)$$

with controls X (size, age) and fixed effects (year, industry, and/or industry $\times$ year). A logit version is also estimated.

	OLS (1)	OLS (2)	Logistic (3)
Tangibility, $t_{i,t-1}$	0.089*** (0.010)	0.090*** (0.010)	0.443*** (0.063)
Volatility, $\delta_{i,t-1}^2$	-0.011*** (0.003)	-0.011*** (0.003)	-0.048** (0.019)
Age	0.048*** (0.005)	0.048*** (0.005)	0.338*** (0.034)
Size	0.066*** (0.002)	0.066*** (0.002)	0.456*** (0.017)
Industry and Year FE	Yes	No	Yes
Industry $\times$ Year FE	No	Yes	No
Observations	77,410	77,410	77,410
R^2	0.198	0.197	0.192

Notes: This table shows the results of regressing an indicator variable taking the value 1 if a firm is public and 0 otherwise on one year lagged firm tangibility, profitability volatility, age, and size. All non-indicator variables are in logs. Industry and year fixed effects are included in the regressions of the first and third columns while industry times year fixed effects are included in the regression of the second column. We adopt the Fama-French 48 industry classification in this exercise. The regression in the first two columns are linear regression while the regression in the third column is a logistic one. Standard errors are heteroskedasticity-robust, clustered at the firm level and reported in brackets. All nominal values are adjusted to 2016 U.S. dollars. We annually winsorize scaled variables without clear upper or lower bounds at the 1% and 99% level. Superscripts *, **, and *** correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.

	CEO Pay (1)	Fixed Share (2)	Incentive Share (3)	Equity Share (4)
Tangibility, $t_{i,t-1}$	-0.342*** (0.023)	0.120** (0.018)	-0.056** (0.023)	-0.076*** (0.025)
Volatility, $\delta_{i,t-1}^2$	0.036*** (0.009)	-0.036*** (0.007)	0.034*** (0.007)	0.044*** (0.007)
Age	0.092*** (0.014)	0.051*** (0.010)	-0.095*** (0.011)	-0.153*** (0.012)
Size	0.464*** (0.007)	-0.215*** (0.005)	0.181** (0.005)	0.176*** (0.005)
Publicly Listed	0.340*** (0.023)	-0.178*** (0.017)	0.356*** (0.031)	0.296*** (0.037)
Industry and Year FE	Yes	Yes	Yes	Yes
Observations	43,757	43,757	39,247	28,054
R^2	0.555	0.266	0.216	0.190

Notes: This table shows the results of regressing the log of the CEO pay, the fixed share, the incentive share, and the equity share on one year lagged firm tangibility, profitability volatility, age, size, and listing status. All non-indicator variables are in logs. Industry and year fixed effects are included in all the regressions, which are all linear. We adopt the Fama-French 48 industry classification in this exercise. Standard errors are heteroskedasticity-robust, clustered at the firm level and reported in brackets. All nominal values are adjusted to 2016 U.S. dollars. We annually winsorize scaled variables without clear upper or lower bounds at the 1% and 99% level. Superscripts *, **, and *** correspond to statistical significance at the 1%, 5%, and 10% levels, respectively.

Quantile	1st	2nd	3rd	4th
Correlation	0.102	0.045	-0.003	-0.251

Notes: This table presents the correlations between the time t GDP growth and the time $t - 1$ number of public domestic listings growth conditional on time $t - 1$ quartiles of VC intensity. The latter is computed as the ratio of VC investments over market capitalization of listed domestic companies. To compute this table we use World Bank WDI data and OECD VC data.

Table 3 results (core coefficients).

- Tangibility is positive and highly significant: $\hat{\beta}_\tau \approx 0.089$ to 0.090 in the LPM columns; in the logit column the coefficient is 0.443 .
- Volatility is negative: $\hat{\beta}_\sigma \approx -0.011$ in the LPM columns; in the logit column -0.048 .

Magnitude: a 10% increase in tangibility implies about a 0.9 percentage-point higher public-listing probability, while a 10% increase in volatility implies about a 0.1 percentage-point lower probability (using the LPM scaling).

Figure 2 (industry scatter). Industries with lower tangibility tend to have higher volatility; private firms are concentrated in the “low- τ , high- σ ” region predicted by (16).

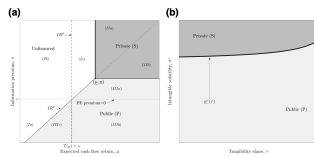


FIGURE 1
Equilibrium firm sorting (a) Theorem 2 Sorting, (b) Corollary 1 Sorting
Notes: Figure 1(a) depicts the equilibrium sorting pattern of Theorem 2 based on firm type's expected cash flow return μ and information premium σ . Figure 1(b) depicts the equilibrium sorting pattern of Corollary 1 based on firm tangibility τ and private cash flow volatility σ^2 . The unshaded region denotes time discounting no financing, the dark shaded region denotes firms receiving private financing, and the light shaded region denotes firms receiving public financing.

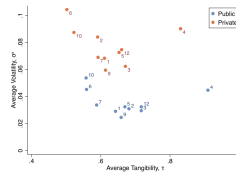


FIGURE 2
Industry evidence about firm listing patterns
Notes: Within industry average tangibility and 3-year earnings volatility conditional on firm listing status (classified by is-sample median). The 12 Fama-French industries considered are 1 run durability, 2 durables, 3 manufacturing, 4 energy, 5 chemicals, 6 business equipment, 7 telecommunications, 8 shops, 9 health, and 12 other. Given our data cleaning filters, the Fama-French industries 8 utilities and 11 money are excluded.

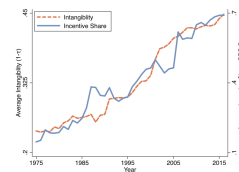


FIGURE 3
Historical CEO pay performance sensitivity and firm tangibility
Notes: This figure depicts the average CEO pay performance sensitivity on the right axis and the average share of tangible assets from 1975 to 2016. The CEO compensation data per 1991 is taken from Frydman and Saks (2010) while post 1991 is from Execucomp. The average tangibility share is constructed using Compustat and Peters and Taylor (2017) data. The sample begins in 1975 since, as explained in Peters and Taylor (2017), this is the first year that the Federal Accounting Standard Board required firms to report R&D.

9.2 CEO pay regressions (Table 4) and time series (Figures 3–4)

Empirical model. For pay levels and pay shares,

$$Y_{it} = \alpha + \beta_\tau \text{Tangibility}_{i,t-1} + \beta_\sigma \text{Volatility}_{i,t-1} + \delta \text{Public}_{it} + \gamma' X_{i,t-1} + \text{FE} + \varepsilon_{it}, \quad (18)$$

where Y_{it} is either $\log(\text{CEO Pay})$ or the fixed/incentive/equity share.

Table 4 results.

- **Pay level:** tangibility loads negatively (-0.342) and volatility positively (0.036); being public adds 0.345 to log pay.
- **Pay composition:**
 - Tangibility increases the fixed share ($+0.120$) and decreases incentive (-0.056) and equity (-0.076) shares.
 - Volatility decreases the fixed share (-0.036) and increases incentive ($+0.034$) and equity ($+0.044$) shares.
 - Public firms have much lower fixed shares and much higher incentive/equity shares.

Interpretation: in precisely the firms where π is large (low tangibility, high volatility), the public-market solution is to load CEO compensation on performance.

Figure 3. The average incentive share of CEO pay rises over time alongside a rise in intangibility.

Figure 4. Rolling regressions show the elasticity of pay with respect to intangibility spikes during the ICT era, consistent with a structural shift in τ (and therefore π).

9.3 International evidence (Figure 5 and Table 5)

Figure 5. Across G7 countries (1996–2014), increases in R&D researcher intensity coincide with declines in the number of domestic listed firms, especially in the U.S.

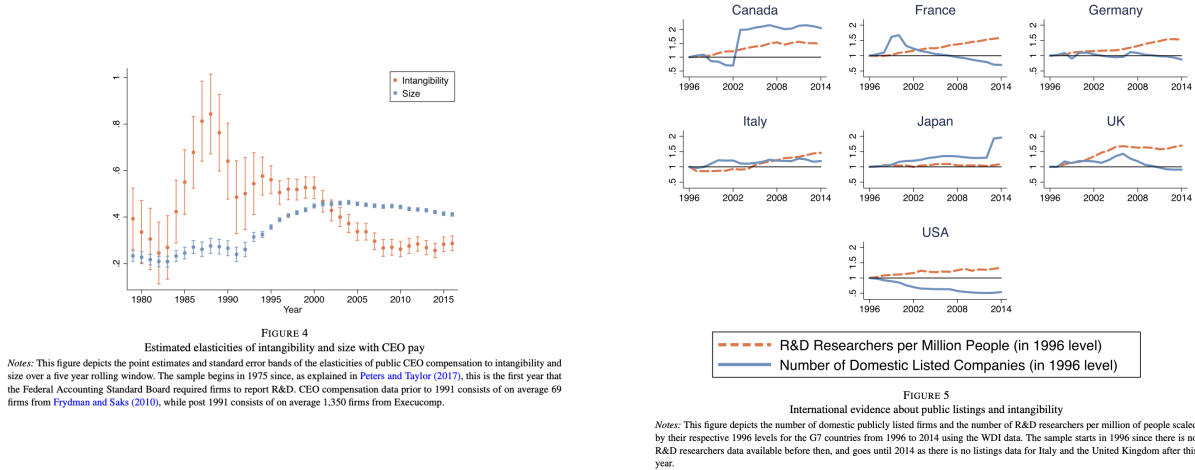


Table 5. Conditional correlations between GDP growth and subsequent listing growth decline with VC intensity:

- 1st quartile VC intensity: $\text{corr} \approx 0.102$,
- 2nd quartile: 0.0845 ,
- 3rd quartile: -0.003 ,
- 4th quartile: -0.251 .

Interpretation: in countries with deeper private-capital markets (high VC intensity), growth is less tied to growth in public listings.

10 Structural estimation: Eqs. (8)–(11) and Table 6

10.1 GMM objective (Eq. (8))

Let δ collect the structural parameters. The estimator solves

$$\hat{\delta} = \arg \min_{\delta} \bar{h}(\delta)' W \bar{h}(\delta), \quad \bar{h}(\delta) = \frac{1}{NT} \sum_i \sum_t h(\delta, \nu_{it}). \quad (19)$$

10.2 Listing moment (Eq. (9))

The paper models listing as

$$L_i = \mathbf{1}\{\nu - \pi_i + \varepsilon_i \geq 0\}, \quad (20)$$

with logistic ε_i . This implies

$$\mathbb{E} \left[\log \left(\frac{\Pr(L_i = 1)}{1 - \Pr(L_i = 1)} \right) \middle| \tau_i \right] = \nu + \beta(1 - \tau_i)^2, \quad (21)$$

where

$$\beta = -\frac{\mathbb{E}[\sigma^2] \psi}{2(\rho + \lambda^{-1})^2}. \quad (22)$$

10.3 Pay-performance sensitivity moment (Eq. (10))

The model implies

$$\frac{\text{Cov}(\omega_t, d_t^y)}{\text{Var}(d_t^y)} = \frac{\mathbb{E}[(1 - \tau)^2 \sigma^2]}{\mathbb{E}[(1 - \tau)^2 \sigma^2 + \tau^2 (\sigma^x)^2]} \cdot \frac{\rho}{\rho + \lambda^{-1}}. \quad (23)$$

10.4 Selection bias (Eq. (11))

Because CEO pay is observed for public firms only, selection biases the pay-based estimate. Ignoring the selection yields a biased expression proportional to

$$\frac{\rho}{\rho + \lambda^{-1}}. \quad (24)$$

The paper corrects using volatility implied by the listing equation.

10.5 Table 6 interpretation (estimated parameters)

Table 6 reports three sets of estimates.

- **Listing-based (col. 1):** $\lambda \approx 7.673$, $\nu \approx 0.664$, $(\sigma^x)^2 \approx 0.003$, $\lambda^x \approx 1.623$, and $\sigma^2 \sim \Gamma(\alpha_0 \approx 2.253, \alpha_1 \approx 0.006)$.

Implications:

- Mean private variance $\mathbb{E}[\sigma^2] = \alpha_0 \alpha_1 \approx 0.0135$, about $4.5\times$ the tangible variance.
- The implied annual AR(1) persistence is $\phi_z = e^{-1/\lambda} \approx 0.878$ vs $\phi_x = e^{-1/\lambda^x} \approx 0.540$. Hence private cash flows are about $1.63\times$ as persistent.

- **CEO-pay-based (col. 2):** $\lambda \approx 5.377$ (implied $\phi_z \approx 0.830$), reflecting selection.
- **Bias-adjusted (col. 3):** $\lambda \approx 8.246$ (implied $\phi_z \approx 0.885$), close to the listing-based estimate.

Big picture: the hidden component is both *riskier* and *more persistent* than tangible cash flows, which rationalizes a high information premium and strong incentive pay in public firms.

TABLE 6
Structural estimation results

Parameter	Firm (1)	CEO (2)	Bias-adjusted CEO (3)
λ —Persistence of private cash flows	7.673 (1.176)	5.377 (1.105)	8.246
ν —Monitoring cost	0.664 (0.036)		
α_0 —Shape of distribution of volatility of private cash flows	2.253 (0.077)		
α_1 —Scale of distribution of volatility of private cash flows	0.006 (0.001)		
$(\sigma^x)^2$ —Volatility of tangible cash flows	0.003 (0.000)		
λ^x —Persistence of tangible cash flows	1.623 (0.109)		

Notes: This table presents the estimates from our structural estimation. Column (1) presents the estimates based on firm listing choice and firm performance and our strategy discussed in subsection 5.2. Column (2) presents the estimate for λ based on CEO pay performance sensitivity and our biased moment discussed in subsection 5.3. Column (3) presents the bias-adjusted estimate of λ from Column (2) using volatility estimates from the firm listing based on equation (11). Bootstrapped standard errors are given in parentheses and are based on 10,000 bootstrap samples.

11 Conclusion

11.1 Summary

This paper argues that the long-run decline in public listings and the simultaneous rise in performance-based CEO compensation can be jointly rationalized by one structural force: **rising exposure to intangible assets that generate persistent hidden information**. Intangible cash flows are difficult for outside investors to verify (and disclosure can be limited by non-separability and imitation risk), so public investors face a persistent hidden-information problem when they finance such firms. In contrast, private investors can acquire information through monitoring and closed-door communication.

The paper builds a market-equilibrium model where public finance must use **costly incentive contracts** to elicit truthful reporting of a persistent hidden cash-flow component, while private finance can **pay a monitoring cost** to observe that component. Because private capital is scarce (or monitoring capacity is limited), the equilibrium features **sorting**: information-intensive firms disproportionately end up privately financed, while relatively transparent firms remain public.

Empirically, the paper validates the sorting logic using data that include both public firms and large private firms, links these patterns to CEO pay composition, and then structurally estimates the persistence and volatility of the hidden component. The estimated magnitudes imply that persistent hidden information can explain a sizable fraction of both the listing decline and the rise of incentive pay.

11.2 Main takeaways

1. **Persistent hidden information is quantitatively important.** The structural estimates imply that the hidden (private-information) component of cash flows is substantially larger and more persistent than the tangible component: the paper reports that the *average* private-information component is about **3.7 times** the tangible component, and that **annual persistence of private information is about 63% higher** than persistence of tangible cash flows (Table 6 and the accompanying discussion).
2. **Persistence magnifies agency distortions beyond what volatility alone would imply.** When private information is persistent, public contracts must discipline a long-lived hidden state. This requires more exposure of CEO compensation to performance innovations and generates an *endogenous* rise in the share of performance-based pay.
3. **A single mechanism links two secular trends: fewer listings and more incentive pay.**

Rising intangibility increases the cost of public finance (via the information premium), pushing more firms into private markets and raising the optimal intensity of incentive pay among firms that remain public.

4. **Counterfactual quantification: rising intangibility can explain a meaningful portion of trends.** The paper reports that average firm intangibility increased by about **55%** from 1980 to 2016 (from **10.31%** to **26.33%**). Holding the model’s estimated parameters fixed, reverting intangibility to the 1980 level would:
 - raise the public-listing probability from **57.22%** to **62.66%** (about a **5.5 pp** increase),
 - reduce the implied information premium from **0.373** to **0.146**, and
 - imply that annual public CEO variable-pay growth would be about **61% lower**.
5. **Interpretation of the listing decline: efficient sorting in the model, but not necessarily welfare-improving in reality.** In the paper’s static financing setting, the decline in public firms can be interpreted as an *efficient market-equilibrium response* to rising informational asymmetries. However, the authors emphasize that richer environments with welfare costs from wealth inequality and from restricted access to private investments could overturn this conclusion. A dynamic extension could generate increasing concentration of wealth among the most able private investors, potentially weakening the link between stock-market performance and domestic economic indicators over time.

11.3 Economic intuition (why everything fits together)

- **Why intangibility matters:** intangible cash flows are harder to verify, so they expand the set of states in which insiders can misreport. When the hidden component is persistent, misreporting today has long-run consequences, so incentives must be stronger and therefore more costly.
- **Why incentive pay rises in public firms:** public investors cannot directly monitor the hidden component, so they substitute toward pay-performance sensitivity. This increases both the level and the performance-based share of compensation, especially among firms that remain public.
- **Why listings fall:** as the cost of public finance rises with the hidden-information premium, privately monitored finance becomes relatively more attractive. Given scarce monitoring capacity, the market sorts the most information-intensive firms into private ownership, shrinking the public-firm sector.
- **Why persistence is the key amplifier:** volatility creates risk, but persistence creates *durability* of misreporting incentives. Therefore persistence raises the informational wedge more than a purely transitory private-information shock would.

11.4 Takeaway

Rising intangibility increases persistent hidden-information frictions, making public finance more incentive-intensive and costly, so equilibrium sorting shifts more firms to private markets while boosting performance-based CEO pay among firms that remain public.